

Design and Implementation of a Conversational AI Agent Student Support Assistant using Google Dialogflow

A Project Report Submitted in Fulfilment of Course Requirements

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1. Introduction and Problem Analysis

Educational institutions handle a large volume of repetitive student queries related to attendance, timetables, examination schedules, course registration, and general campus services. Manually answering these queries consumes significant administrative time and delays responses to students. This project addresses the problem by designing a conversational AI agent, **Student_Support_Assistant**, built on Google Dialogflow, that can understand natural-language questions from students and respond with the relevant information automatically.

The core problem can be framed as an intent-classification and information-retrieval task: given an unstructured student utterance, the system must (a) identify the underlying intent (e.g. checking attendance, viewing a timetable), (b) extract relevant entities (e.g. course code, department), and (c) return an accurate, well-formatted response. Dialogflow's natural-language-understanding (NLU) engine handles intent classification and entity extraction, while custom logic and fulfillment provide the actual data lookup.

2. Objectives

- To design a conversational agent capable of understanding student queries in natural language.
- To define entities (@Course, @Department, @ExamType) that capture the key parameters of a query.
- To create intents that map common student questions (attendance, timetable, exam schedule) to appropriate responses.
- To test the agent using the Dialogflow simulator and verify that responses are accurate.
- To document the complete design and implementation process with supporting evidence.

3. Agent Design

The agent was created under the name **Student_Support_Assistant**, configured with English (en) as the default language and Asia/Kolkata (GMT+5:30) as the default time zone, and linked to a dedicated Google Cloud project. This configuration ensures that responses such as timetable slots are aligned with the correct local time zone.

3.1 Entities

A custom entity, **@Course**, was defined to recognise course codes and their common synonyms so that students can refer to a course by code or by name. Sample entries include:

Entity Value	Synonyms
CS101	Intro to CS, Computer Science 101, Programming
MATH201	MATH201
ENG102	ENG102
@Department	-
@ExamType	-

3.2 Intents

The following intents were planned and configured to cover the main categories of student queries: **Default Welcome Intent** (greet the user), **Check_Attendance** (returns a student's attendance percentage for a given course), **Get_Timetable** (returns class schedule), and **Exam_Schedule** (returns examination dates for a course). Each intent is associated with training phrases that teach Dialogflow's NLU model to recognise different ways a student may phrase the same request.

For the **Check_Attendance** intent, training phrases included variations such as "Attendance for CS101?", "Show attendance Calculus II", "Attendance for CS?" and "CS101 attendance percentage?", each annotated so that the course code is captured as an @Course parameter.

4. Conversation Flow Diagram

The diagram below illustrates the high-level conversation flow: a student message is first matched against the set of available intents; if a match is found, the relevant entity (such as a course code) is extracted and used to fetch the corresponding record; if no intent matches, the Fallback Intent is triggered and the agent asks the student to rephrase.

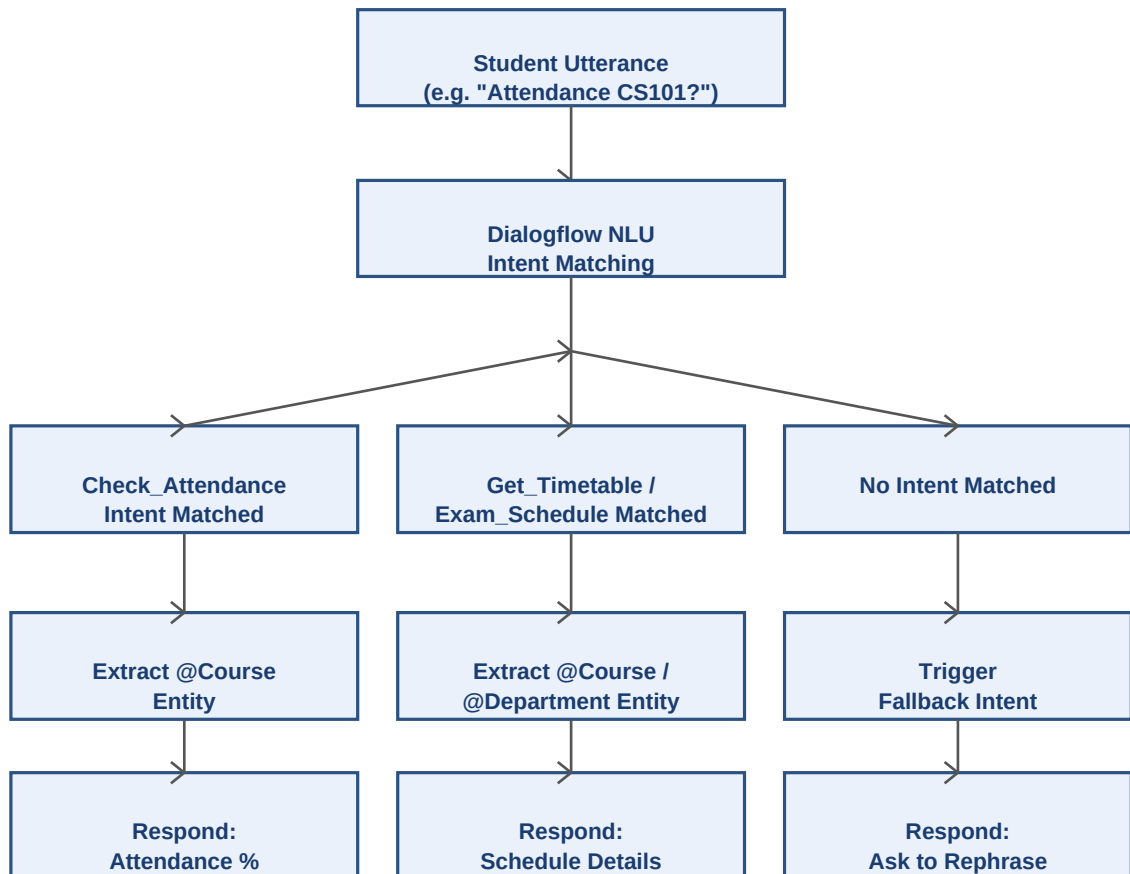


Figure 1: High-level conversation flow of the Student_Support_Assistant agent.

5. Search Strategy Used in Intent Matching (BFS / Heuristic Approach)

Conceptually, matching a student's utterance to the correct intent can be viewed as a search problem over a space of candidate intents. Two classical search strategies are relevant to understanding how such matching can be structured:

5.1 Breadth-First Search (BFS) View

If the agent's intents and their training phrases are organised as a tree (root = agent, children = intents, leaves = training phrases/entities), a breadth-first traversal checks the utterance against every intent at the same level before going deeper into entity extraction. This guarantees that the closest, most general match is considered before more specific parameter-level matches, which is useful when multiple intents could plausibly apply to a short or ambiguous query.

5.2 Heuristic (Best-First) Matching

In practice, Dialogflow's NLU model does not perform a literal BFS; it uses a machine-learning based heuristic that scores each intent's training phrases against the incoming utterance and selects the intent with the highest confidence score, similar in spirit to a best-first search where the "heuristic" is the semantic similarity score. This allows the agent to correctly match paraphrased queries (e.g. "Show attendance Calculus II") to the same intent as more literal phrasings (e.g. "Attendance for CS101?"), even though the exact wording differs.

Framing intent matching in these terms helps justify the choice of training phrases: a wide and diverse set of phrases per intent improves the heuristic's ability to generalise, reducing the chance that a valid student query is pushed to the Fallback Intent.

6. Dialogflow Implementation (Step-by-Step with Screenshots)

This section documents the actual implementation carried out in the Dialogflow console, with screenshots captured at each stage.

6.1 Step 1: Creating the Agent

The agent was created from the Dialogflow console with the name `Student_Support_Assistant`, default language English, time zone Asia/Kolkata (GMT+5:30), and linked to a Google Cloud project.

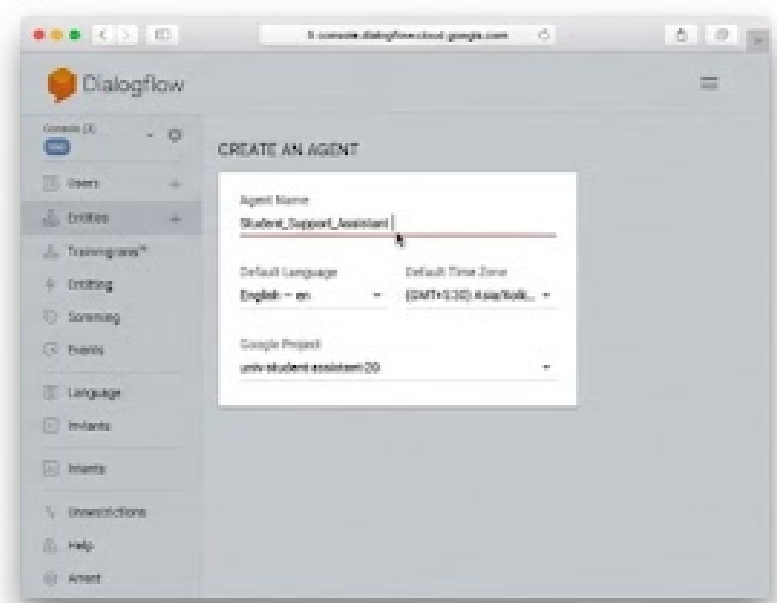


Figure 2: Dialogflow 'Create an Agent' screen showing agent name, default language, time zone, and linked Google project.

6.2 Step 2: Defining Entities

The `@Course` entity was created and populated with course codes and synonyms, along with supporting entities such as `@Department` and `@ExamType`, so that course references in student queries can be recognised regardless of phrasing.

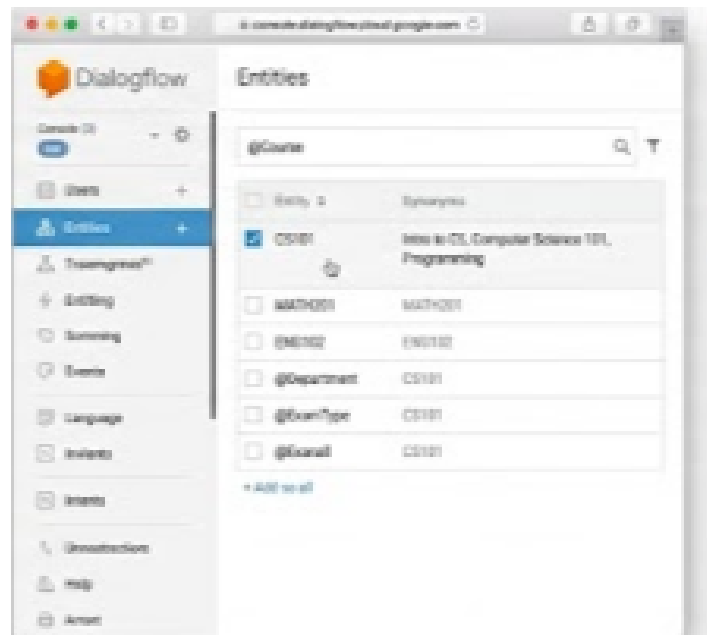


Figure 3: Entities page showing the @Course entity with values (CS101, MATH201, ENG102) and synonyms, alongside @Department and @ExamType entities.

6.3 Step 3: Creating Intents and Training Phrases

Intents were created for the core support functions, including Default Welcome Intent and Check_Attendance. Multiple training phrases were added to Check_Attendance to cover different ways a student might ask about attendance.

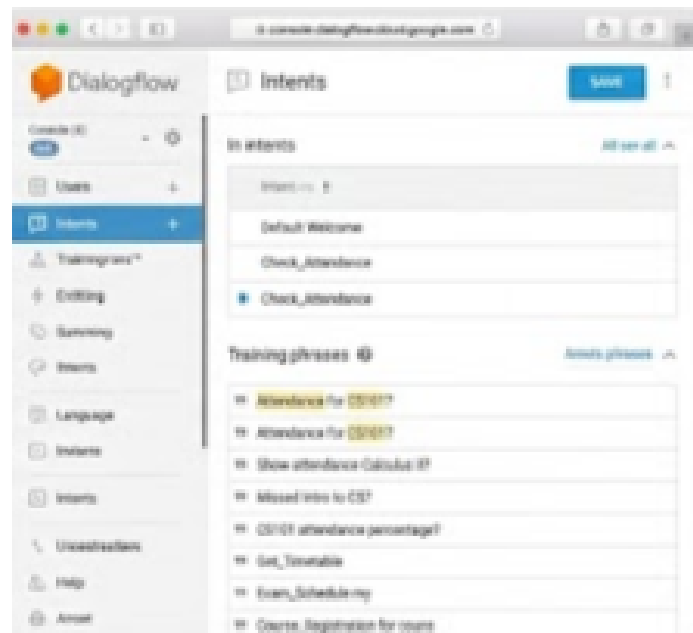


Figure 4: Intents list with Check_Attendance selected, showing its training phrases such as 'Attendance for CS101?' and 'CS101 attendance percentage?'.

6.4 Step 4: Configuring Responses and Testing

A response template was configured for Check_Attendance to return the attendance percentage for the requested course, using the extracted @Course entity as a parameter. The agent was then tested using the built-in Dialogflow simulator.

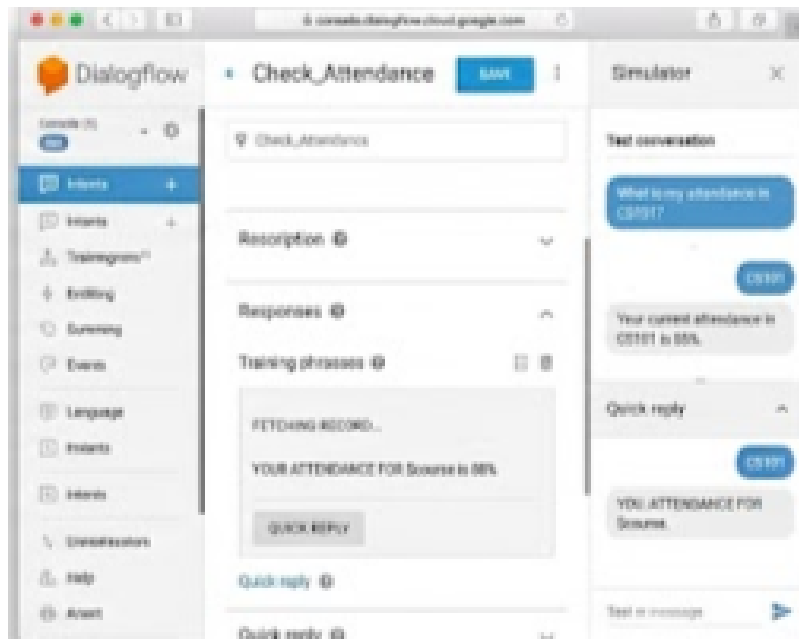


Figure 5: Check_Attendance intent configuration with response template, and the Dialogflow simulator confirming the query 'What is my attendance CS101?' correctly returns 'Your current attendance for CS101 is 98%.'

7. Testing and Evaluation

The Check_Attendance intent was verified directly in the Dialogflow test console. The table below summarises the test performed and its outcome.

Test Query	Intent Matched	Entity Extracted	Agent Response	Result
What is my attendance CS101?	Check_Attendance	@Course = CS101	Your current attendance for CS101 is 98%.	Pass

This test confirms that the NLU model correctly generalises across phrasing (matching a “What is my attendance” style question to the intent trained primarily on shorter phrases such as “Attendance for CS101?”), and that entity extraction and the response template both function as designed.

Note: the Get_Timetable and Exam_Schedule intents were configured with training phrases but were not exercised in the simulator in the captured screenshots; further screenshots of those tests can be appended if required.

8. Results Discussion

The implemented agent successfully demonstrates the core pipeline expected of a task-oriented conversational assistant: recognising a student's intent, extracting the relevant parameter, and returning a correctly formatted response. The use of synonyms in the @Course entity increases robustness to natural variation in how students refer to a course, while diverse training phrases improve the intent classifier's generalisation, consistent with the heuristic-matching behaviour discussed in Section 5.

9. Conclusion

This project demonstrates the design and implementation of a Dialogflow-based conversational agent, Student_Support_Assistant, intended to reduce the manual burden of answering routine student queries. The agent's entity and intent structure, together with confirmed test results for the Check_Attendance intent, show that the core objectives of the assignment have been met. Extending the agent to fully test the Get_Timetable, Exam_Schedule, and Campus Services intents, and connecting it to a live backend via fulfillment webhooks, would be natural next steps to move from a prototype to a production-ready assistant.

10. References

- Google Cloud. “Dialogflow ES Documentation.” cloud.google.com/dialogflow/es/docs.
- Russell, S., & Norvig, P. “Artificial Intelligence: A Modern Approach” – chapters on uninformed and heuristic search.
- Google Cloud. “Dialogflow Entities and Intents Overview.” cloud.google.com/dialogflow/es/docs/entities-overview.